Avery Mattoon | New York, NY | (781) 491-6228 | averyrmattoon@gmail.com | LinkedIn | GitHub

M.S. Computer Science candidate with a background in economics, research experience at a central bank, and IR/treasury experience at a life science tools company. Skilled in building Python and R pipelines for economic research, NLP on financial text, and analysis of multi-million-record datasets. Available for full-time roles from May 2027.

EDUCATION

Emory University, Laney Graduate School — **Atlanta, GA**
Master of Science: Computer Science — *Expected May 2027*
- Cumulative GPA: 4.0/4.0

Emory University, College of Arts and Sciences — **Atlanta, GA**
Bachelor of Arts: Economics and Computer Science, Spanish minor — *May 2026*
- Cumulative GPA: 3.9/4.0
- Completed academic terms abroad at the University of Salamanca (2023) and DIS Copenhagen (2024)

Relevant Coursework: Econometrics, Computing Methods in Economics, Data Science for Economists, Machine Learning, Natural Language Processing, Data Mining, Database Systems, Artificial Intelligence, Information Visualization

PROFESSIONAL EXPERIENCE

Central Reserve Bank of Peru — **Lima, Peru**
Research Assistant, Macroeconomic Modeling — *June 2025 – August 2025*
- Developed an end-to-end NLP pipeline testing corporate-filing sentiment as a GDP indicator; found that bank CEO-letter sentiment tracks same-year growth ($r \approx 0.5$, 2000–2024), a coincident rather than leading signal
- Built a Selenium web scraper to collect up to 25 years of annual reports for 266 firms, replacing 440+ hours of manual collection time
- Automated text extraction from collected filings into an analysis-ready dataset, using OCR to digitize the ~30% that were scanned
- Developed sentiment indices using a custom Spanish financial dictionary, isolating executive summaries as the most signal-bearing sections

Waters Corporation — **Milford, MA**
Finance Intern, Investor Relations & Treasury — *May 2024 – August 2024*
- Modeled 14 cross-currency swaps ($550M+) hedging Euro net investment in Bloomberg (SWPM, MARS) and built a live valuation dashboard, giving treasury on-demand valuations in place of monthly statements from counterparty banks
- Automated Smartsheet data collection for the annual global insurance renewal (~$3.5B total insured value, 90 facilities in 35 countries), saving 5 hours of manual work per week; prepared exposure data for broker meetings
- Prepared regular macro and peer briefings for the Head of IR, synthesizing Bloomberg market data, competitor filings, and analyst Q&As

Emory University — **Atlanta, GA**
Instructor, DATASCI 100 Lab (Introductory Data Science in R) — *January 2026 – Present*
- Teach R-based statistics and data analysis lab sections for an introductory course of 160 undergraduate students
- Write lab quizzes, homework, and practice worksheets, and manage 4 undergraduate teaching assistants

RELEVANT PROJECTS

H-1B Economic Impact Analysis (PostgreSQL, Python, Next.js) – live dashboard | code repository — *2025 – 2026*
- Built a PostgreSQL pipeline for 7.4M certified H-1B applications (FY2009–2024), reconciling 16 years of shifting DOL formats
- Deployed a live Next.js dashboard stress-testing a $100K per-filing fee (~$51B/year upper bound), with IT services bearing 37%+ of the cost
- Caught an FY2019 error that recorded 300K+ filings at employer HQs instead of worksites, skewing state-level YoY changes by up to 170 percentage points

Era Bias in Music Popularity Prediction (Python) – code repository — *2025*
- Compared 7 regression models (linear, tree-based, kNN) predicting popularity of 580K+ Spotify tracks (1921–2020), validated with 5-fold CV
- Showed models trained on one era fail on others: worse than a simple average in 19 of 20 cross-era tests (Random Forest)
- Found a hidden confound: guessing each era's average popularity matched the top model's R^2 (0.34), so its accuracy was mostly era

SKILLS

- **Programming Languages:** Python (pandas, NumPy, scikit-learn, spaCy, matplotlib), R (tidyverse), SQL (PostgreSQL), JavaScript (D3)
- **Tools & Platforms:** Excel, Bloomberg Terminal (SWPM, MARS, Excel add-in), Git/GitHub, Jupyter, Google Colab, Selenium
- **Methods:** Econometrics, regression, machine learning, statistics, NLP, sentiment analysis, web scraping, PDF parsing, data visualization
- **Certifications & Languages:** Bloomberg Market Concepts; Spanish (professional working proficiency)

INTERESTS

- Hiking (all-time favorite: W trek in Patagonia, solo); NYT crossword (recently conquered Saturday); Geography (GeoGuessr, MapTap)